

(1) Research objectives, research methods, comparison with previous studies, originality, and relevance to international research exchange activities

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The objective of this study is to conduct an ex ante assessment of how and under what conditions “digital policy bundles”—including digital agricultural advisory services, digital finance, e-commerce, and logistics support—can strengthen the resilience of rural households and regional food systems in the Solo/Surakarta region of Central Java, Indonesia. Droughts associated with climate change, fluctuations in agricultural prices, disruptions in logistics, and information gaps in rural areas are major factors that destabilize food security. Digital technologies may improve access to market information, financial services, sales channels, and transport coordination. However, their effects do not automatically emerge simply through technological introduction. When digital tools are linked with existing local institutions such as cooperatives, mutual-help practices, trust-based relationships, and collective decision-making, they may enhance the capacity of households and communities to respond to shocks. Conversely, if individual-based applications bypass existing cooperative mechanisms, they may weaken local solidarity, increase inequalities within villages, or create new forms of dependency. The central purpose of this study is therefore to identify whether digital policy bundles complement or substitute for existing community-based resilience mechanisms, and to clarify desirable policy combinations before full-scale implementation.

Previous studies have accumulated substantial knowledge on digital agriculture, financial inclusion, agricultural information services, food security, and climate adaptation. However, many of these studies tend to focus on the average effects of single interventions, adoption intention, productivity, or income. They have not sufficiently examined how local institutions, market access, ecological risks, household resources, and digital readiness interact with one another to produce resilience. In addition, while ex post evaluations after policy implementation are relatively common, fewer studies assess in advance the potential effectiveness, side effects, and distributional consequences of policy bundles. This study addresses these gaps by treating digitalization not merely as a technical solution, but as a policy instrument embedded in social institutions, markets, ecological conditions, and local governance structures.

The originality of this study lies in three main points. First, it applies the perspective of institution–technology complementarity. Digital technologies are not assumed to generate resilience independently; rather, their effectiveness is understood as depending on how they are combined with existing local institutions, including cooperatives, mutual-help systems, and collective action. Second, the study conceptualizes food-system resilience as an integrated outcome consisting of food security, shock buffering and recovery, market continuity, ecological stability, and adaptive capacity. This makes it possible to evaluate resilience not only in terms of short-term income or productivity, but also in terms of stability, recovery, and equity under shocks. Third, the study combines a policy vignette experiment with fuzzy-set Qualitative Comparative Analysis (fsQCA). This approach enables the research to identify not only whether a digital policy bundle is effective on average, but also which combinations of institutional, economic, ecological, household-level, and policy conditions are sufficient for achieving high resilience.

The study will be conducted in the Solo/Surakarta region, where peri-urban and rural agricultural systems, irrigated and drought-prone areas, and traditional institutions and emerging digital services coexist. Approximately 6 to 8 villages will be selected to reflect variation in ecological, market, and institutional conditions. A short but intensive field survey will be conducted with approximately 120 to 160 rural households, combined with a policy vignette experiment and village-level key informant interviews. The household survey will collect data on household characteristics, land, income, assets, food security, dietary diversity, shock exposure, recovery time, coping strategies, participation in cooperatives and mutual-help networks, trust, market access, mobile phone use, digital finance, and agricultural or commercial app use. Through these data, the study will capture both household-level adaptive capacity and village-level resilience mechanisms.

The policy vignette experiment will present respondents with hypothetical policy scenarios that combine different elements, such as digital agricultural advisory services, digital finance, e-commerce, logistics support, and delivery modes through cooperatives or directly to individual households. Respondents will be asked about their likelihood of adoption, expected food security under shocks, expected stability of production, and risk of falling into distress coping strategies. This design allows the study to evaluate potential responses to digital policy bundles before actual large-scale implementation. Village-level data will also be collected through key informants on institutional strength, collective action, infrastructure, irrigation, roads, storage, drought exposure, and market linkages.

For the analysis, the study will construct an Integrated Food-System Resilience Index (IFSR) based on five dimensions: food security adequacy, shock buffering and recovery, market continuity, ecological stability, and adaptive capacity. The interaction between digital policy bundles and local institutions will then be estimated to determine whether digital interventions complement, substitute for, or operate independently of existing local resilience mechanisms. Furthermore, fsQCA will be used by defining each household or village under each policy scenario as a case. Conditions will include local institutions, household resources, market access, ecological conditions, digital readiness, and the type of policy bundle. Through this configurational analysis, the study will identify sufficient combinations of conditions that generate high, robust, and equitable food-system resilience.

Because the research involves household surveys and interviews with human participants, ethical considerations will be integrated from the beginning of the project. Participants will be informed of the research purpose, voluntary participation, the right to withdraw, anonymization procedures, and the scope of data use before consent is obtained. Personal identifiers will be separated from survey data, access to data will be restricted, and any publication or data sharing will be conducted in a form that does not identify individuals or villages.

The relevance of this study to international research exchange activities is also significant. Prior to this application, collaboration with Indonesian researchers has already been developed through academic exchange and discussions on food-system resilience, rural institutions, local wisdom, and policy responses in Indonesia. In particular, Indonesian counterpart researchers visited Tokyo University of Science in October 2025 and presented their research

on how Indonesia should build food-system resilience, including the Pancasila economic paradigm, government intervention, local wisdom, food security challenges, and lessons from Japan's food-system strategy. These exchanges clarified the strong complementarity between the Japanese side's expertise in digital policy evaluation, resilience measurement, and analytical methods, and the Indonesian side's expertise in rural resilience, food security, local institutions, agricultural economics, and stakeholder engagement.

Building on these exchanges, the present project aims to develop a sustained bilateral research network under the JSPS–BRIN framework. The Japanese side will mainly contribute to theoretical framing, indicator construction, ex ante evaluation design, econometric analysis, fsQCA, and international publication. The Indonesian side will provide essential knowledge of local institutions, agricultural distribution systems, ecological risks, policy processes, field-site selection, research permissions, questionnaire localization, enumerator training, and feedback to local stakeholders. By integrating these complementary strengths, the project will not be a one-time field survey, but a collaborative process of knowledge co-production. It will contribute to both academic research and practical policy design by clarifying how digital transformation can strengthen food-system resilience without undermining the institutional strengths of rural communities.

(2) Preparatory Status and Implementation Plan

1. Previous exchanges and negotiation process with the counterpart researchers

This project is planned on the basis of an emerging collaborative relationship between the Japanese team and Indonesian researchers. From 13 to 21 September 2025, preliminary meetings toward joint research with Universitas Indonesia were held at the Noda Campus of Tokyo University of Science. The shared theme was strengthening rural resilience for regional revitalization, and the two sides exchanged views on the research objectives, target areas, survey design, data collection methods, and directions for future dissemination. In these discussions, the central issues were household responses to shocks, local institutions, agriculture, logistics, market access, and dialogue with policy stakeholders. The Japanese side presented the theoretical framework, indicator design, and analytical methods, while the Indonesian side examined local institutional contexts, regional administration, candidate research sites, and possibilities for collaboration with relevant stakeholders.

In October 2025, Indonesian-side researchers visited Tokyo University of Science and held further discussions with the Japanese side. During this visit, Dr. Palupi Lindiasari Samputra of Universitas Indonesia and Prof. Dr. Agr. Sc. Ernoiz Antriandarti of Universitas Sebelas Maret presented their research on food-system resilience in Indonesia, including the Pancasila economic paradigm, local wisdom, government intervention, food security challenges, and lessons from Japan's food-system strategy. This visit served as an important opportunity to deepen mutual understanding of the research agenda, clarify the relevance of Indonesian local institutions and food-system resilience to the proposed project, and discuss possible directions for a JSPS–BRIN bilateral joint research application. The exchange also helped identify how Japanese perspectives on digital policy evaluation and Indonesian expertise in

rural resilience, food security, and local institutional systems can be integrated into a concrete joint research design.

The expertise of the counterpart researchers is directly relevant to this project. The Japanese side has strengths in informatics, social systems analysis, policy evaluation, statistical analysis, and configurational analysis, and will take the lead in designing the digital policy bundles, developing resilience indicators, conducting empirical analysis, and preparing joint publications. The Indonesian side has strengths in strategic and global studies, national resilience, regional revitalization, food-system resilience, rural institutions, agribusiness, agricultural economics, environmental and social challenges, and stakeholder research. In particular, the Indonesian team's previous studies on food insecurity, local wisdom, dryland farm households, and climate-related vulnerability provide an essential empirical and conceptual basis for this project. This complementarity of expertise enables the proposed research to evaluate digital technologies not as stand-alone tools, but as policy instruments embedded in local institutions, market access, ecological risks, and implementation processes.

In this application, the collaborative relationship developed through these exchanges will be further advanced into a bilateral joint research project with BRIN. Based on earlier discussions on Indonesian rural resilience, regional policy, data collection and analysis, and stakeholder engagement, the proposed project focuses on the Solo/Surakarta region, where peri-urban and rural agricultural systems, irrigated and drought-prone areas, and traditional institutions and emerging digital services coexist. With the counterpart researchers, the Japanese side will finalize the English project title, target area, Indonesian-side principal investigator, research period, BRIN-side application procedure, research permits, ethics review, participating researchers, data management rules, and methods for returning research results to local stakeholders. To ensure consistency between the Japanese and Indonesian applications, the names of the principal investigators and the English project title will be matched in both submissions. Online meetings and email consultations will be used before submission to confirm that the BRIN-side application is completed by the required deadline. After the project begins, the Indonesian side's expertise will also be used to finalize the Indonesian-language questionnaire, back-translation, recruitment of field enumerators, explanation procedures for target villages, and coordination with village heads, cooperatives, agricultural extension officers, and local governments.

2. Annual division of responsibilities, exchange plan, and travel/hosting plan

The project period is planned for two years, from 1 April 2027 to 31 March 2029. The Japanese side will be responsible for developing the theoretical framework integrating institution–technology complementarity, configurational resilience, and ex ante policy evaluation; constructing the Integrated Food-System Resilience Index (IFSR); designing the policy vignette experiment; conducting econometric analysis and fsQCA; and preparing joint academic papers and policy briefs. The Indonesian side will be responsible for providing knowledge of local institutions, markets, and ecological conditions; selecting target villages; obtaining survey permissions; localizing the questionnaire; training enumerators; conducting key informant interviews; coordinating with local research institutions, local governments, and cooperatives; and organizing feedback activities in Indonesia. Both sides will jointly conduct online monthly meetings, a kick-off meeting, fieldwork, data checking sessions, analysis workshops,

young researcher seminars, and final result meetings. The project will be implemented as a genuinely joint research process, from research design to dissemination.

The first half of FY2027 will be positioned as the stage for building the research foundation. The Japanese side will review the literature, prepare the survey and vignette instruments, define the components of the IFSR, prepare ethics review materials, design the data management plan, and develop the basic analytical code. The Indonesian side will handle BRIN-side procedures, compile candidate villages, coordinate with local governments and village-level actors, identify field enumerators, and review the Indonesian-language questionnaire. An online kick-off meeting will be held in the first half of the year to confirm the research questions, division of responsibilities, survey schedule, risk management, research ethics, and scope of data sharing. Japanese early-career researchers will participate in this process whenever possible, so that they can learn the practical procedures of international joint research from the preparatory stage.

The second half of FY2027 will be the main period for fieldwork and face-to-face exchange. Japanese researchers will make a short-term visit to Indonesia. After research meetings with the Indonesian counterpart researchers and, where appropriate, BRIN-related stakeholders in Jakarta, the two sides will jointly conduct enumerator training, a pilot survey, the main household survey, and key informant interviews in the Solo region. The survey will cover approximately 6 to 8 villages and 120 to 160 households across different ecological, market, and institutional conditions. The Indonesian side will lead coordination with village heads, farmer organizations, cooperatives, agricultural extension officers, and local governments, and will take the lead in explaining the study in the local language, obtaining informed consent, and ensuring survey quality. The Japanese side will be responsible for sampling design, vignette allocation, data entry rules, and consistency checks of the survey instruments. After the fieldwork, both sides will jointly conduct data cleaning, missing-value checks, translation checks, and preliminary analysis, and will identify any additional interview items that may be needed.

The first half of FY2028 will be devoted to joint analysis and interim result sharing. The Japanese side will construct the IFSR, estimate the interaction between digital policy bundles and local institutions, conduct fsQCA to identify configurations of sufficient conditions, and perform robustness checks. The Indonesian side will verify whether the analytical results are consistent with the realities of local institutions, market structures, ecological conditions, and policy implementation contexts, and will consider how to provide feedback to villages and policy stakeholders. Online analysis meetings will be held approximately once a month. Early-career researchers will participate in analytical coding, data management, interpretation of results, and preparation of English manuscripts. If necessary, Japanese researchers will make an additional short-term visit to Indonesia to conduct supplementary interviews and hold an interim feedback session.

The second half of FY2028 will be the stage for integrating outcomes and planning future development. Both sides will prepare international joint papers, policy briefs, survey methodology materials, and a plan for a subsequent comparative research project. A final results meeting will be organized for BRIN, Universitas Indonesia, Universitas

Sebelas Maret, local governments, cooperatives, and other relevant stakeholders. If the Indonesian-side researchers can utilize BRIN, institutional, or researcher mobility funding, a joint analysis meeting and young researcher workshop may be held in Japan. If a visit to Japan is not feasible, equivalent opportunities for joint discussion will be secured online. The JSPS budget will be allocated mainly to the Japanese researchers' travel to Indonesia, domestic travel, survey preparation, and dissemination activities, ensuring that travel expenses account for at least 50 percent of the total annual commissioned budget as required for joint research. By involving early-career researchers in the stages of survey preparation, fieldwork, analysis, and dissemination, the project will foster practical capacity in international joint research, fieldwork, mixed methods, and policy communication. After completion, the project will be developed into comparative research on Southeast Asian food-system resilience, research on regional agricultural digital transformation and disaster-resilient food systems in Japan, and future larger-scale joint research applications.

Because the project involves surveys and interviews with human participants, ethics review, informed consent documents, anonymization procedures, and data-sharing rules will be confirmed by both sides from the first year. The research will comply with relevant laws, institutional regulations, and local practices in both countries. In the event of bad weather, delayed research permits, changes in target villages, or travel restrictions, multiple candidate villages will be prepared in advance, and the project will shift, if necessary, to online interviews or supplementary data collection by the Indonesian counterpart researchers. These measures will strengthen the feasibility and continuity of the research plan.